

# Sanghyun Lee

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## RESEARCH INTERESTS

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I am broadly interested in diffusion models, with a current focus on discrete diffusion models and unified multimodal models. My research particularly focuses on improving the efficiency, scalability, and reasoning capabilities of diffusion language models, and on extending them toward unified multimodal models that understand and generate across modalities within a single model.

On the inference side, I am interested in test-time scaling, designing inference algorithms that effectively scale computation at generation time. On the training side, I am interested in modeling, developing architectures and formulations that enable efficient and scalable discrete diffusion learning.

## EXPERIENCE

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### Krafton

Research Intern, Foundation Research Team

2025.06 – 2026.04

Seoul, South Korea

## EDUCATION

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### Korea Advanced Institute of Science and Technology (KAIST)

Integrated M.S.-Ph.D., Kim Jaechul Graduate School of AI

Advisor: Prof. Seungryong Kim

2025.09 – Present

Daejeon, South Korea

### Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Electronic Engineering, Minor in Mathematical Sciences

2019.03 – 2025.09

Daejeon, South Korea

## PUBLICATIONS

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### International Conference

1. **Sanghyun Lee**, Chunsan Hong, Seungryong Kim, Jonghyun Lee, Jongho Park, Dongmin Park. “**Looped Diffusion Language Models**”  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2026. [\[link\]](#)
2. Chunsan Hong\*, **Sanghyun Lee\***, Chieh-Hsin Lai, S. Hayakawa, Y. Takida, Yuki Mitsufuji, Seungryong Kim, J.C. Ye. “**Understanding and Accelerating the Training of Masked Diffusion Language Models**”  
*International Conference on Machine Learning (ICML) workshop*, 2026. [\[link\]](#)
3. **Sanghyun Lee**, Seungryong Kim, Jongho Park, Dongmin Park. “**Lookahead Unmasking Elicits Accurate Decoding in Diffusion Language Models**”  
*International Conference on Machine Learning (ICML)*, 2026. [\[link\]](#)
4. Chunsan Hong, **Sanghyun Lee**, J.C. Ye. “**Unifying Masked Diffusion Models with Various Generation Orders and Beyond**”  
*International Conference on Machine Learning (ICML)*, 2026. *Spotlight Paper (Acceptance rate < 2.2%)*. [\[link\]](#)
5. **Sanghyun Lee**, Sunwoo Kim, Seungryong Kim, Jongho Park, Dongmin Park. “**Effective Test-Time Scaling of Discrete Diffusion through Iterative Refinement**”  
*International Conference on Machine Learning (ICML) workshop*, 2026. [\[link\]](#)
6. Donghoon Ahn\*, Jiwon Kang\*, **Sanghyun Lee**, Jaewon Min, Minjae Kim, Wooseok Jang, Hyoungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha, Kyong Hwan Jin, Seungryong Kim. “**A Noise Is Worth Diffusion Guidance**”  
*International Conference on Learning Representations (ICLR)*, 2026. [\[link\]](#)
7. Donghoon Ahn\*, Jiwon Kang\*, **Sanghyun Lee**, Minjae Kim, Jaewon Min, Wooseok Jang, Sangwu Lee, Sayak Paul, Susung Hong, Seungryong Kim. “**Fine-Grained Perturbation Guidance via Attention Head Selection**”  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [\[link\]](#)

## RESEARCH AREAS

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- Discrete diffusion models

- Unified multimodal models
- Test-time scaling
- Probabilistic modeling
- Efficient from-scratch pretraining

## SKILLS

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- **Programming and Systems:** Python, PyTorch, C/C++, CUDA
- **Large-Scale Training Experience:**
  - Billion-scale language model and unified multimodal model from-scratch pretraining
  - Training experience on multi-GPU clusters
- **Academic Knowledge:** Diffusion Models, Reinforcement Learning Theory, Information Theory, Linear Algebra, MC Methods, Optimization Theory, Measure Theory, Real Analysis